

Younos Hashem

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EDUCATION

Stony Brook University <i>Ph.D. in Physics</i>	Stony Brook, NY 2032 (expected)
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Georgia Institute of Technology <i>B.S. in Physics</i> GPA: 4.0	Atlanta, GA 2026
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RESEARCH EXPERIENCE

Georgia Institute of Technology <i>Undergraduate Research Assistant</i> Advisor: Dr. Jesse McDaniel	Atlanta, GA September 2024 – Present
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- Developed a Python package to perform QM/MM calculations on covalently bonded systems with 1000s of atoms.
- Investigated behavior of silica–water interfaces using quantum mechanical and molecular dynamics computational methods in a high-performance computing environment.
- Analyzed data to turn simulation results into knowledge of surface chemistry.
- Performed extensive literature searches to understand and model the complex electrostatics of periodic systems.

Georgia Institute of Technology, Quantum Computing Association <i>Project Developer</i>	Atlanta, GA September 2025 – Present
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- Implemented decomposition of multi–logical qubit states based on fast Walsh–Hadamard transform.
- Studied representations of logical quantum states with error correction.

Kennesaw State University <i>Undergraduate Research Assistant</i> Advisor: Dr. Martina Kaledin	Kennesaw, GA May – August 2023
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- Co-authored a paper on potential energy surface fitting. Paper was published in the *Journal of Chemical Theory and Computation*.
- Took lead of a group of three students on a project involving the theoretical study of peptide isomers.
- Characterized by computational methods the potential energy surface of formamide and its isomers.

Kennesaw State University <i>Research Assistant, First-Year Scholars Program</i> Advisor: Dr. Jeremy Gulley	Kennesaw, GA October 2022 – April 2023
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- Ran numerical simulations for a work on laser-induced damage to nanoscale semiconductors.
- Developed simplified numerical models as proof-of-concept.

TEACHING EXPERIENCE

Georgia Institute of Technology

Atlanta, GA

Teaching Assistant: Principles of Physics II, Optics, Linear Algebra

Spring 2025

Led student problem-solving sessions, delivered lectures, and worked with students one-on-one

PUBLICATIONS

“Fitting Potential Energy Surfaces by Learning the Charge Density Matrix with Permutationally Invariant Polynomials”. Y. Hashem, K. Foust, M. Kaledin, A. Kaledin, *J. Chem. Theory Comput.* 2023, 19, 17, 5690–5700.

PAPERS UNDER REVIEW

“LogicalQ: A toolkit for quantum circuit development with generalized quantum error mitigation, detection, and correction”. R. Devkota, B. Hagan, N. Heffner, Y. Hashem, N. Song, A. Deshpande, F. Makinde, R. Yu, A. Petrusinskaia, Z. Song, D. Lloyd George, L. Lampert, *JOSS* (under review).

AWARDS AND GRANTS

President’s Undergraduate Research Award

Summer 2025, Spring 2026

Research grant (\$3,500)

RESEARCH PRESENTATIONS

“Damage to Nanoscale Semiconductor Wires from Ultrashort Laser Pulses”

April 2023

KSU Undergraduate Research Symposium

Advisor: Dr. Jeremy Gulley

SKILLS

Programming Python (advanced), C (intermediate), FORTRAN (novice)

Simulation codes Psi4, VASP, OpenMM

Computational tools Git, Slurm, Bash

LEADERSHIP EXPERIENCE

Society of Physics Students, Georgia Tech Chapter

Atlanta, GA

Treasurer

May 2025 – Present

Managed budget and billing process worth \$1000s of dollars per semester for an organization with dozens of members. Planned events with 100+ attendees to raise hundreds of dollars.